

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

Kopiereg voorbehou

Examination

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Module EBN122
Elektrisiteit en Elektronika
14 November 2014

Module EBN122
Electricity and Electronics
14 November 2014



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	90	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 minute <i>180 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		27 (Antwoordblad uigesluit / <i>Answer sheet excluded</i>)	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Die Engelse teks van hierdie vraestel sal voorkeur geniet met betrekking tot die interpretasie van die inhoud.
The English text of this question paper will be given precedence in the interpretation of the content.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Polêre vorm:** $A \angle \theta$; **Reghoekige vorm:** $a + jb$
Polar form: $A \angle \theta$; **Rectangular form:** $a + jb$
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Interne Eksaminator(e): <i>Internal examiner(s):</i>	J. Joubert
	J.D. le Roux
Eksterne eksaminator: <i>External examiner:</i>	W. Odendaal

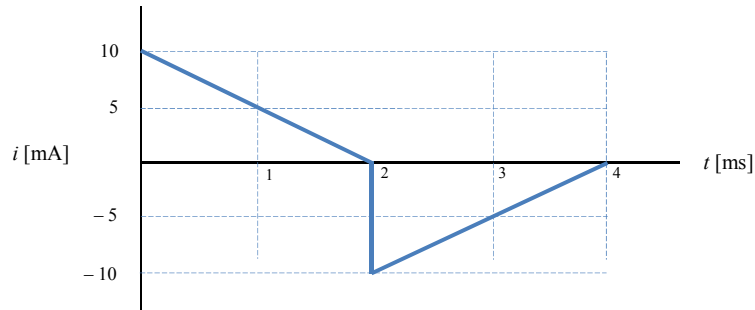
VRAAG/QUESTION 1 [15]

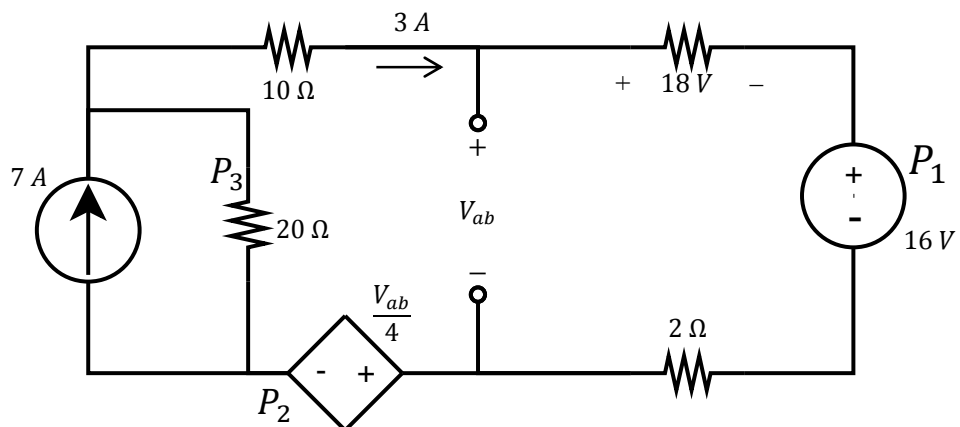
Vraag 1.1 [4]

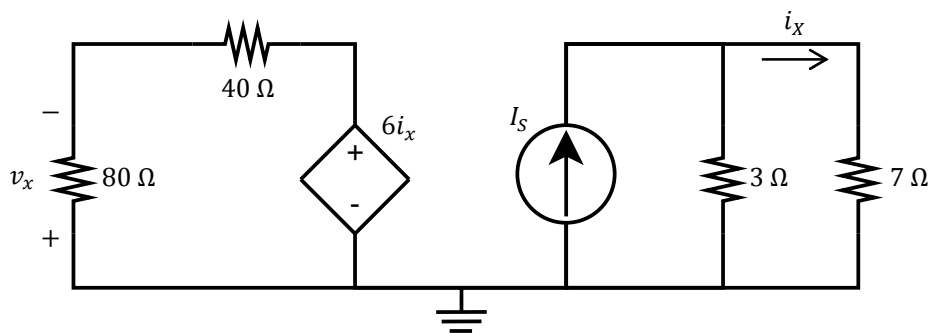
- a) 'n Ketel is gekoppel aan 'n bron van 150 V en benodig 10 A vir 5 minute om water te kook. Bepaal die energie verbruik van die ketel in kWh.
- b) Die stroom wat deur 'n $10\ \Omega$ resistor vloei word in die onderstaande grafiek aangedui. Bepaal die lading wat deur die resistor tussen $t = 1\text{ ms}$ en $t = 4\text{ ms}$ vloei.

Question 1.1 [4]

- a) A kettle is powered by 150 V and draws a current of 10 A for 5 minutes to boil water. Determine the energy used by the kettle in kWh.
- b) The current flowing through a $10\ \Omega$ resistor is shown in the graph below. Determine the charge through the resistor between $t = 1\text{ ms}$ and $t = 4\text{ ms}$.



Vraag 1.2 [3]Bepaal P_1 , P_2 and P_3 .**Question 1.2 [3]**Determine P_1 , P_2 and P_3 .

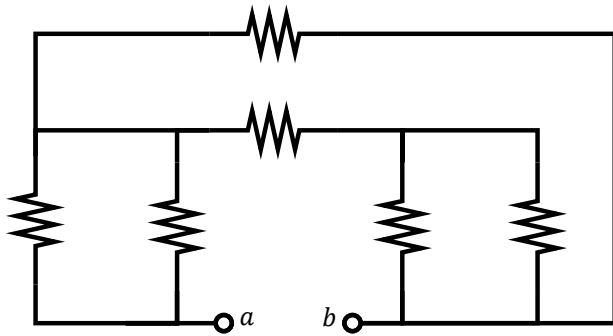
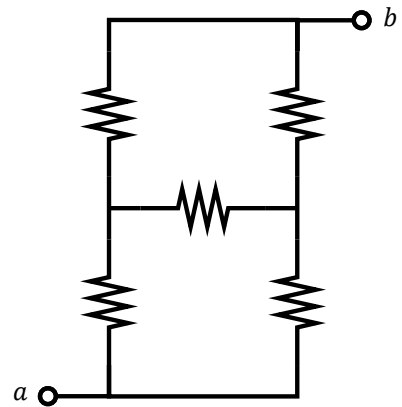
Vraag 1.3 [4]Indien $i_x = 3$ A, bepaal v_x en I_S .**Question 1.3 [4]**If $i_x = 3$ A, determine v_x and I_S .

Vraag 1.4 [4]

Bepaal die ekwivalente weerstand van onderstaande twee kringbane, ten opsigte van terminale a en b .
Aanvaar alle weerstande het 'n waarde van $30\ \Omega$.

Question 1.4 [4]

Determine the equivalent resistance of the two circuits below with respect to terminals a and b . Assume all resistors have a value of $30\ \Omega$.

Kringbaan / Circuit 1**Kringbaan / Circuit 2**

VRAAG/QUESTION 2 [30]

Vraag 2.1 [4]

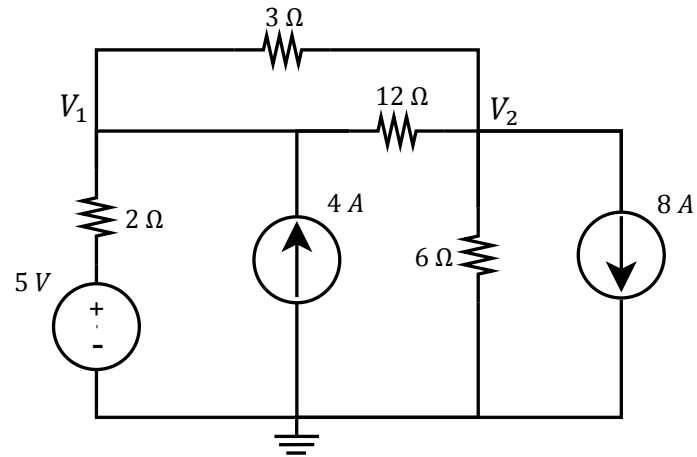
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned}aV_1 + bV_2 &= 78 \\cV_1 + dV_2 &= -96\end{aligned}$$

Question 2.1 [4]

Use nodal analysis to set up two equations of the following form:

$$\begin{aligned}aV_1 + bV_2 &= 78 \\cV_1 + dV_2 &= -96\end{aligned}$$



Vraag 2.2 [4]

Gebruik lusanalyse om 'n vergelyking van die volgende vorm op te stel:

$$ei_1 + fi_2 = 30 V$$

$$A - B = 4 mA$$

(Vervang A met i_1 of i_2 , en B met i_1 of i_2 sodat die vergelyking waar is.)

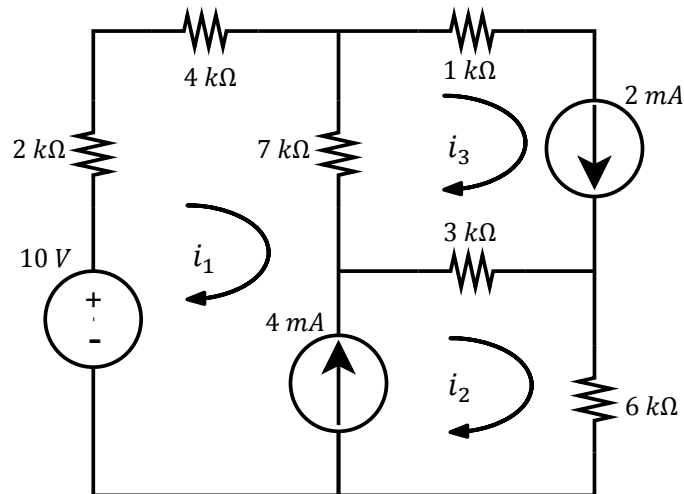
Question 2.2 [4]

Use mesh analysis to set up two equations of the following form:

$$ei_1 + fi_2 = 30 V$$

$$A - B = 4 mA$$

(Substitute A with i_1 or i_2 , and B with i_1 or i_2 so that the equation is true.)



Vraag 2.3 [4]

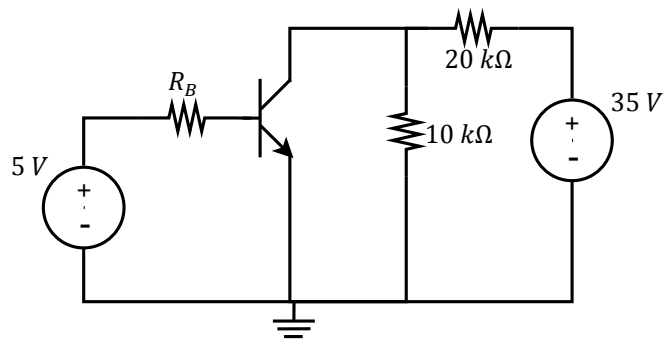
Gegee: $\beta = 100$; $V_{BE} = 0.7 \text{ V}$. Indien $I_B = 10 \mu\text{A}$, bepaal:

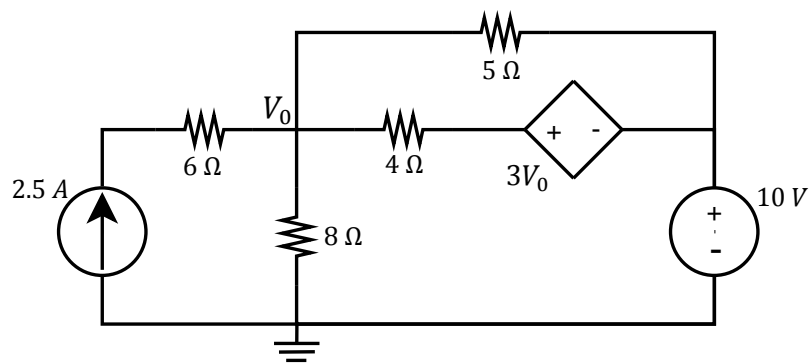
- a) R_B
- b) V_{CE}

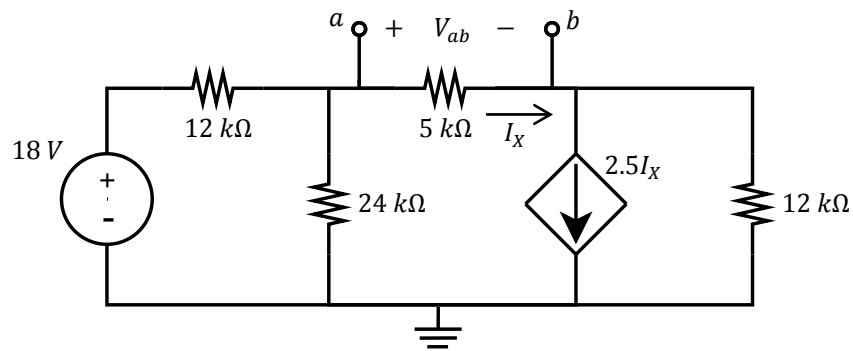
Question 2.3 [4]

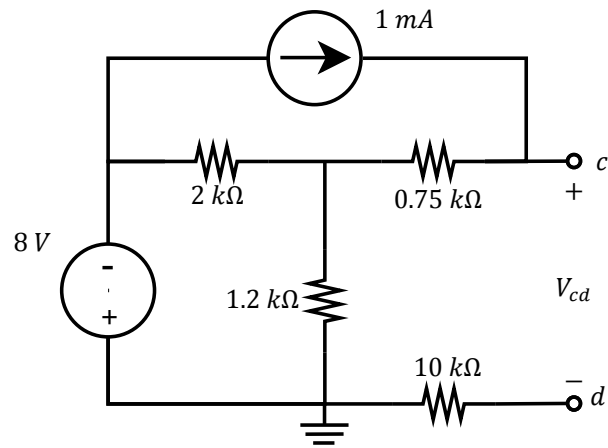
Given: $\beta = 100$; $V_{BE} = 0.7 \text{ V}$. If $I_B = 10 \mu\text{A}$, calculate:

- a) R_B
- b) V_{CE}



Vraag 2.4 [2]Bepaal die node spanning V_0 .**Question 2.4 [2]**Determine the nodal voltage V_0 .

Vraag 2.5 [2]Maak gebruik van brontransformatie en bepaal I_X .**Question 2.5 [2]**Use source transformation to determine I_X .

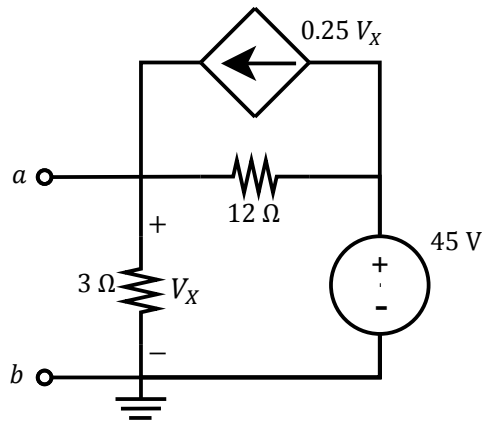
Vraag 2.6 [2]Bepaal die bydrae van die 1 mA bron tot V_{cd} .**Question 2.6 [2]**Calculate the contribution of 1 mA source to V_{cd} .

Vraag 2.7 [4]

Bepaal die Thevenin ekwivalent V_{TH} en R_{TH} met betrekking tot terminale a - b .

Question 2.7 [4]

Determine the Thevenin equivalent V_{TH} and R_{TH} with respect to terminals a - b .

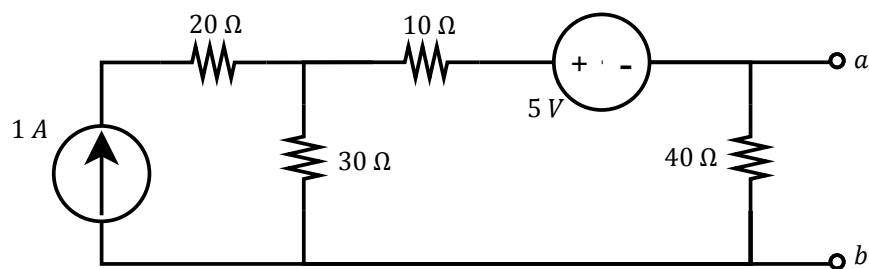


Vraag 2.8 [4]

Bepaal die Norton ekwivalent I_N en R_N met betrekking tot terminale a en b .

Question 2.8 [4]

Determine the Norton equivalent current I_N and resistance R_N with respect to terminals a and b .



Vraag 2.9 [4]

Vir onderstaande kringbaan is $I_N = 0.5 \text{ A}$ en $R_N = 5 \Omega$.

a) Indien $R_L = 5 \Omega$, bepaal P_L (die drywing verkwis deur die lasweerstand).

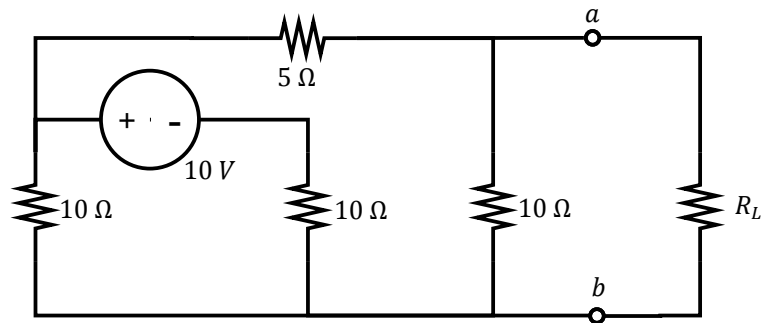
b) Indien $R_L = 10 \Omega$, bepaal P_L .

Question 2.9 [4]

For the circuit below $I_N = 0.5 \text{ A}$ and $R_N = 5 \Omega$.

a) If $R_L = 5 \Omega$, determine P_L (the power dissipated by the load resistor.)

b) If $R_L = 10 \Omega$, determine P_L .



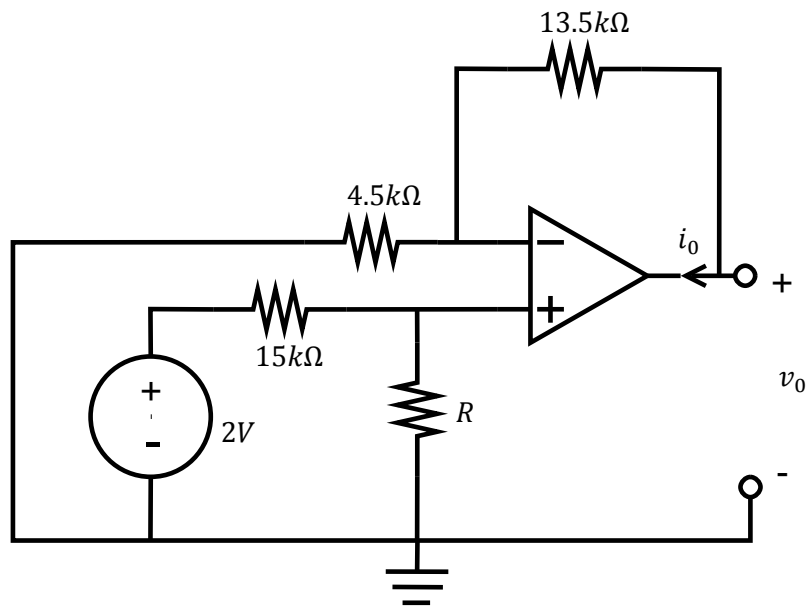
VRAAG/QUESTION 3 [22]

Vraag 3.1 [4]

- a) Bereken v_o as $R = 15\text{ k}\Omega$.
b) Bereken i_o as R verwyder word uit die stroombaan.

Question 3.1 [4]

- a) Determine v_o if $R = 15\text{ k}\Omega$.
b) Determine i_o if R is removed from the circuit.

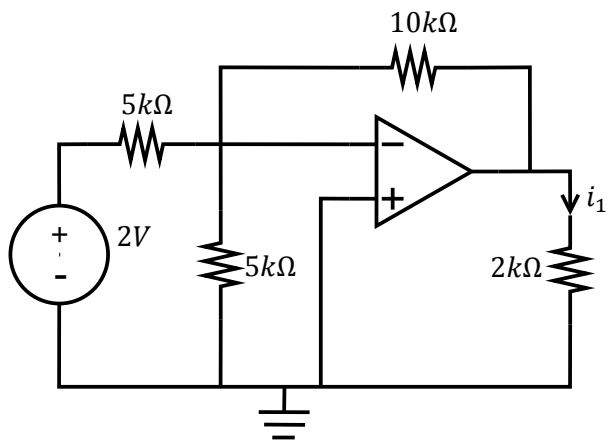


Vraag 3.2 [4]

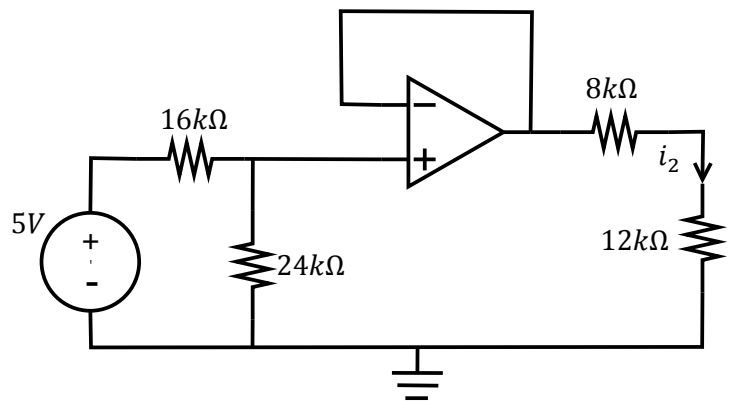
- a) Bereken i_1 (skryf die antwoord in mA).
b) Bereken i_2 (skryf die antwoord in mA).

Question 3.2 [4]

- a) Determine i_1 (write the answer in mA).
b) Determine i_2 (write the answer in mA).



(a)



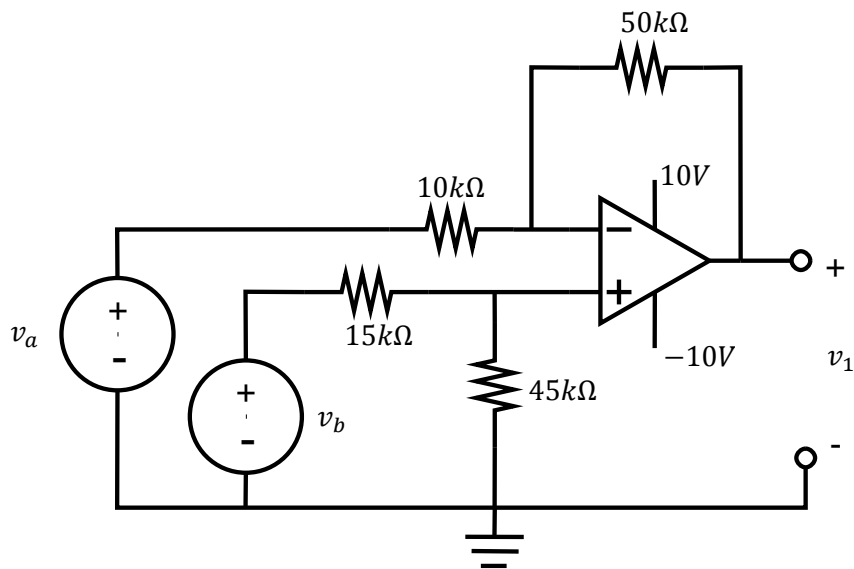
(b)

Vraag 3.3 [4]

- a) Bereken v_I as $v_a = 0.5\text{V}$ en $v_b = 2\text{V}$.
b) Aanvaar $v_b = 0\text{V}$. Wat is die minimum waarde wat v_a kan wees voordat die OV in versadiging ingaan.

Question 3.3 [4]

- a) Determine v_I if $v_a = 0.5\text{V}$ and $v_b = 2\text{V}$.
b) Assume $v_b = 0\text{V}$. What is the minimum value v_a can be before the op-amp goes into saturation.

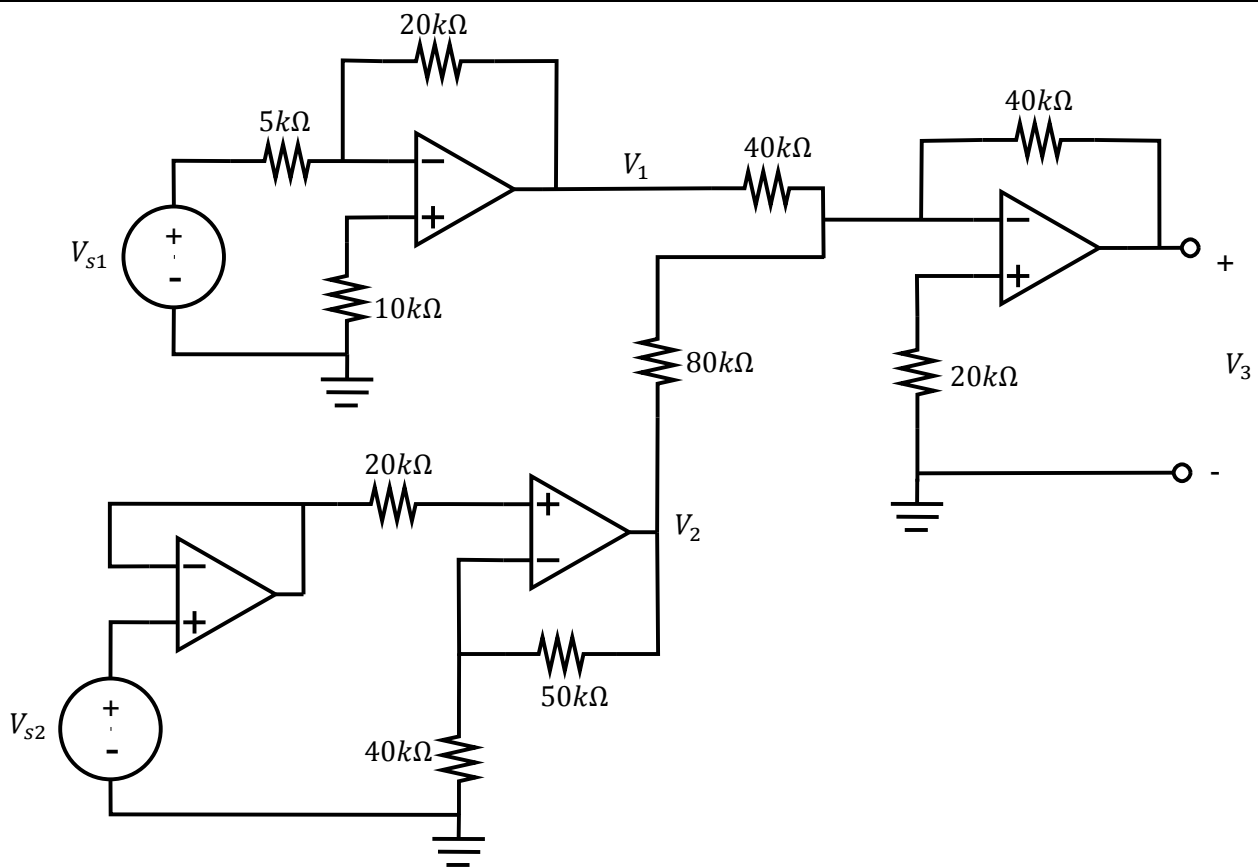


Vraag 3.4 [6]

- a) Bereken V_1/V_{s1} .
- b) Bereken V_2/V_{s2} .
- c) Bereken V_3 as $V_1 = 1\text{V}$ en $V_2 = 2\text{V}$.

Question 3.4 [6]

- a) Determine V_1/V_{s1} .
- b) Determine V_2/V_{s2} .
- c) Determine V_3 if $V_1 = 1\text{V}$ and $V_2 = 2\text{V}$.

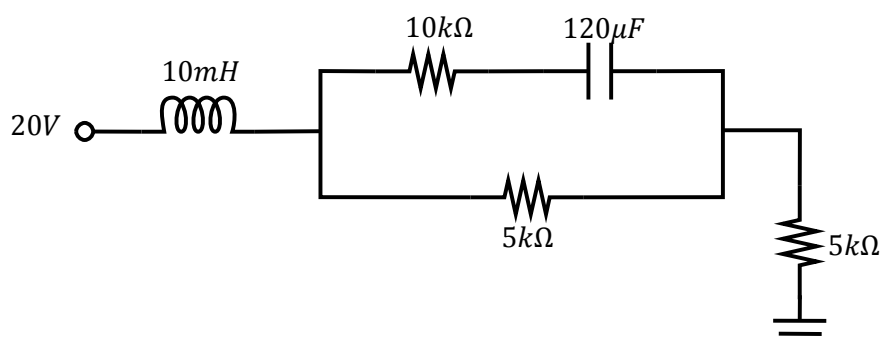


Vraag 3.5 [2]

Aanvaar die baan is in stabiele-toestand en bereken die energie wat gestoor is in die $120\ \mu F$ kapasitor.

Question 3.5 [2]

Assume steady-state conditions and determine the energy stored in the $120\ \mu F$ capacitor.

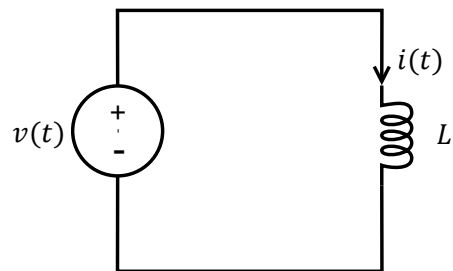


Vraag 3.6 [2]

Bereken die induktorstroom by $t = 1 \text{ ms}$ as $v(t) = 157.08 \sin(250\pi t) \text{ V}$, $L = 50 \text{ mH}$, en $i(0) = -5 \text{ A}$.

Question 3.6 [2]

Find the inductor current at $t = 1 \text{ ms}$ if $v(t) = 157.08 \sin(250\pi t) \text{ V}$, $L = 50 \text{ mH}$, and $i(0) = -5 \text{ A}$.



VRAAG/QUESTION 4 [23]

Vraag 4.1 [4]

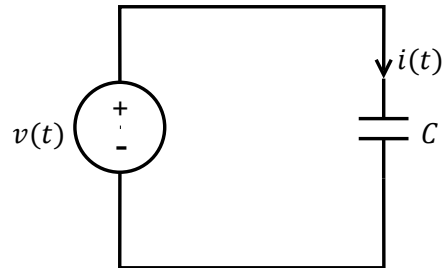
‘n 4 kHz sinusvormige spanning met nul fasehoek en ‘n maksimum amplitude van 2.5 mV (van die vorm $A\cos(\omega t)$) word aangelê oor die terminale van ‘n kapasitor. Die maksimum amplitude van die stroom deur die kapasitor is $125.67\text{ }\mu\text{A}$.

- Wat is die fasehoek van die stroom, in grade.
- Wat is die kapasitansie van die kapasitor in μF .

Question 4.1 [4]

A 4 kHz sinusoidal voltage with zero phase angle and a maximum amplitude of 2.5 mV (of the form $A\cos(\omega t)$) is applied across the terminals of a capacitor. The maximum amplitude of the current through the capacitor is $125.67\text{ }\mu\text{A}$.

- What is the phase angle of the current, in degrees.
- What is the capacitance of the capacitor in μF .

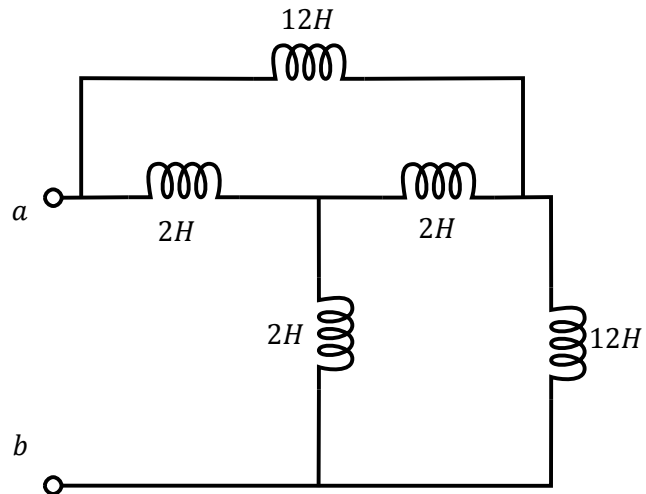


Vraag 4.2 [2]

Bereken die impedansie Z_{ab} as $\omega = 10$ rad/s. Skryf die antwoord in reghoekige vorm.

Question 4.2 [2]

Find the impedance Z_{ab} if $\omega = 10$ rad/s. Write the answer in rectangular form.



Vraag 4.3 [4]

- a) Skryf die volgende tyd-domein spanning as 'n fasor in polêre vorm: $v(t) = 250 \cos(377t + 30^\circ) - 250 \sin(377t - 150^\circ) V$.
- b) Vereenvoudig die volgende uitdrukking en skryf die antwoord in polêre vorm:

$$\bar{I} = \frac{(3 - j4)^*(-3 + j4)}{\sqrt{4} \angle 60^\circ + j \sin(30^\circ)}$$

Question 4.3 [4]

- a) Write the following time-domain voltage as a phasor in polar form: $v(t) = 250 \cos(377t + 30^\circ) - 250 \sin(377t - 150^\circ) V$.
- b) Simplify the following expression and write the answer in polar form:

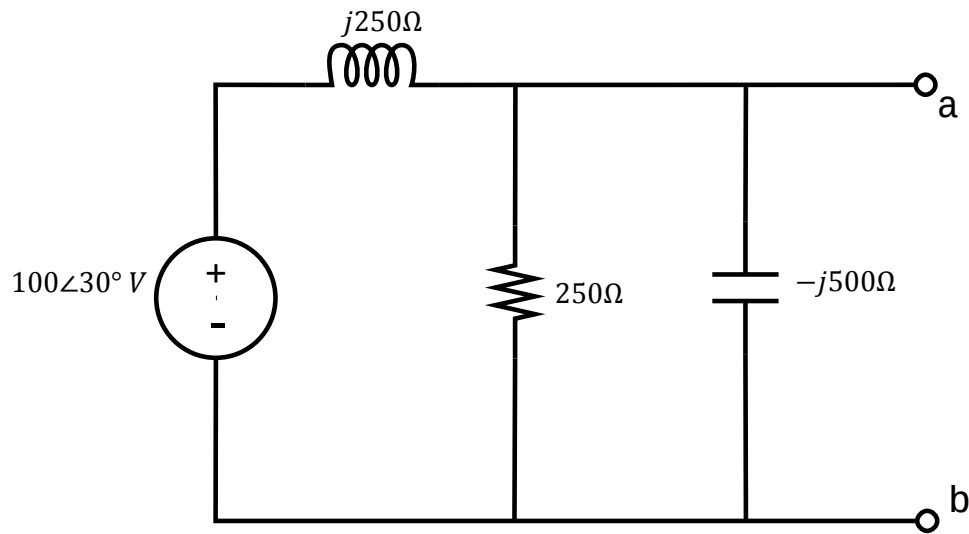
$$\bar{I} = \frac{(3 - j4)^*(-3 + j4)}{\sqrt{4} \angle 60^\circ + j \sin(30^\circ)}$$

Vraag 4.4 [4]

Bepaal die Norton ekwivalent tov terminale a-b. Skryf $\bar{I}_N$ in polêre vorm, en Z_N in reghoekige vorm.

Question 4.4 [4]

Find the Norton equivalent wrt terminals a-b. Write $\bar{I}_N$ in polar form, and Z_N in rectangular form.

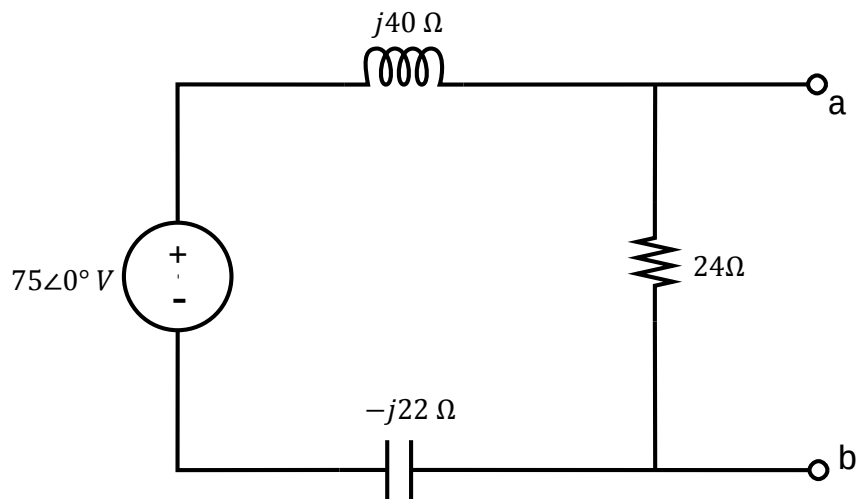


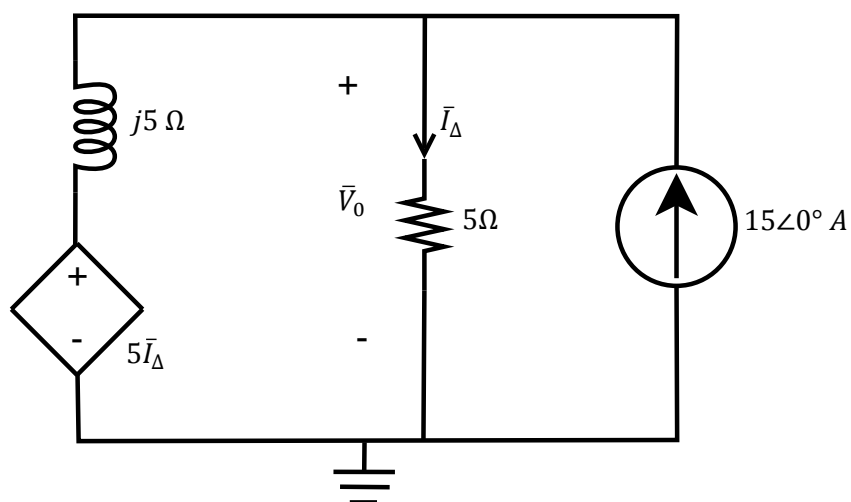
Vraag 4.5 [4]

Bepaal die Thevenin ekwivalent tov terminale a-b. Skryf $\bar{V}_{Th}$ in polêre vorm, en Z_{Th} in reghoekige vorm.

Question 4.5 [4]

Find the Thevenin equivalent wrt terminals a-b. Write $\bar{V}_{Th}$ in polar form, and Z_{Th} in rectangular form.



Vraag 4.6 [2]Bepaal $\bar{V}_0$. Skryf die antwoord in polêre vorm.**Question 4.6 [2]**Determine $\bar{V}_0$. Write the answer in polar form.

Vraag 4.7 [3]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$(a_1 + jb_1)\bar{I}_1 + (a_2 + jb_2)\bar{I}_2 = 6\angle -90^\circ$$

$$(a_3 + jb_3)\bar{I}_1 + (a_4 + jb_4)\bar{I}_2 = 2\angle -90^\circ$$

Question 4.7 [3]

Use mesh analysis to set up two equations of the following form:

$$\bar{I}_1 - \bar{I}_2 = A\angle\theta^\circ$$

$$(a_1 + jb_1)\bar{I}_1 + (a_2 + jb_2)\bar{I}_2 = 2\angle 0^\circ$$

